

Michael Yang

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EDUCATION

Carleton University

Ottawa, ON

Bachelor of Computer Science, Honours in AI and ML, Co-op Option

Sept. 2023 – Apr 2028

- **Relevant Coursework:** Abstract Data Types and Algorithms, Discrete Structures II, Introduction to Systems Programming, Introduction to Software Engineering

TECHNICAL SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, HTML/CSS, Lua, C#, Go

Frameworks/Libraries: React.js, Node.js, Pytest, Flask, TailwindCSS, Lang Chain, Express.js, streamlit, agno, unittest

Developer Tools: Git, Docker, MongoDB, Linux, Github, Jupyter, Confluence, Gitlab, Jira, TestRail, Prometheus, Grafana

EXPERIENCE

Software Developer Co-op

Summer 2026

JSI Telecom

Ottawa, ON

- Incoming Summer 2026

Software Developer Co-op

Sept 2025 – Dec 2025

JSI Telecom

Ottawa, ON

- Built **LLM agents** using **Python**, **LanceDB**, and the **Agno** framework and validated model outputs with **DeepEval** to ensure consistency and quality.
- Achieved **90%+ test coverage** by implementing robust unit and integration tests with **pytest**, ensuring high system reliability, and documented code in **MKDocs**.
- Developed a custom wrapper for **HTTP** and **gRPC** clients that monitors outgoing requests with **Prometheus** and built observability dashboards using **Grafana** to proactively monitor traffic, latency, error rates, and **LLM usage metrics**.

Software Developer Co-op - Incubation

June 2025 – Aug 2025

Nokia

Ottawa, ON

- Developed a **Python-based simulation tool** to model and validate a proof of concept for proactively managing **network traffic flow** in **5G networks** and packet distribution management.
- Implemented **parameterized unit tests** with **Pytest** to enhance code robustness, scalability, and long-term maintainability.
- Collaborated in a cross-functional **SCRUM** team, actively participating in daily standups and sprint planning to ensure alignment between development goals and project deliverables.

PROJECTS

Matchify | *TypeScript, React, Flask, MongoDB, Node.js, Spotify API, OpenAI API, Python*

- Spearheaded the back-end development using **Python** and **Flask**, implementing database management and **API Integrations**.
- Used **Spotify API** to implement **user authentication** to analyze users top songs and artists, generating personalized music taste profiles.
- Integrated **AI-powered recommendations** using **OpenAI API** for song discovery and potential user connections.

Alsk | *JavaScript, React, LangChain, TailwindCSS, Pinecone, Express.js, MongoDB, Tavily, Node.js*

- Developed a Healthcare chat bot within a **MERN Stack** web application allowing users to chat about healthcare-related topics.
- Built a **Retrieval-Augmented Generation (RAG)** system using **LangChain**, **Tavily**, and **Pinecone** to enhance the chatbot's ability to provide accurate and context-aware responses.
- Designed a simple, user-friendly UI using **React** and **TailwindCSS**, and built **RESTful APIs** with **Express.js** to enable efficient client-server communication.